

Data control language

Data control language (DCL) is used to access the stored data. It is mainly used for revoke and to grant the user the required access to a database. In the database, this language does not have the feature of rollback.

- It is a part of the **structured query language (SQL)**.
- It helps in controlling access to information stored in a database. It complements the **data manipulation language (DML)** and the **data definition language (DDL)**.
- It is the simplest among three commands.
- It provides the administrators, to remove and set database permissions to desired users as needed.
- These commands are employed to grant, remove and deny permissions to users for retrieving and manipulating a database.

GRANT Command

It is employed to grant a privilege to a user. **GRANT command** allows specified users to perform specified tasks

Syntax

GRANT privilege_name on objectname to user;

Here,

- privilege names are SELECT,UPDATE,DELETE,INSERT,ALTER,ALL
- objectname is table name
- user is the name of the user to whom we grant privileges

REVOKE Command

It is employed to remove a privilege from a user. **REVOKE** helps the owner to cancel previously granted permissions.

Syntax

REVOKE privilege_name on objectname from user;

Here,

- privilege names are SELECT,UPDATE,DELETE,INSERT,ALTER,ALL
- objectname is table name
- user is the name of the user whose privileges are removing

Example

GRANT SELECT, UPDATE ON employees TO Bhanu

Explanation ? Firstly, to give the permissions to user, we have to use **GRANT** command. The privileges are **SELECT** because to view the records and **UPDATE** to modify the records. The *objectname* is table name which is *Employee*. The user name is "bhanu".

REVOKE SELECT, UPDATE ON employees TO Bhanu

Explanation ? Firstly, to revoke the permissions to user, we have to use **REVOKE** command. The privileges Need to revoke are **SELECT** because to view the records and **UPDATE** to modify the records. The *objectname* is table name which is *Employee*. The user name is "Bhanu".

TCL Commands

Transaction Control language is a language that manages transactions within the database. It is used to execute the changes made by the DML statements.

Transaction Control Language (TCL) Commands are:

- **Commit** – It is used to save the transactions in the database.
- **Rollback** – It is used to restore the database to that state which was last committed.
- **Begin** – It is used at the beginning of a transaction.
- **Savepoint** – The changes done till savpoint will be unchanged and all the transactions after savepoint will be rolled back.

Example

Given below is an example of the usage of the TCL commands in the database management system (DBMS) –

```
BEGIN TRANSACTION
```

```
UPDATE employees
```

```
SET empname='bob'
```

```
WHERE empid='001'
```

```
UPDATE employees
```

```
SET empname ='bob'
```

```
WHERE city='hyderabad'
```

```
IF @@ROWCOUNT=5
```

```
    COMMIT TRANSACTION
```

```
ELSE
```

```
    ROLLBACK TRANSACTION
```

In the above example after we begin the transaction, we are trying to update the employee's name with some value of id. If we affect five rows with our first query then, it will COMMIT transaction else It will be ROLLBACK.

Difference between Commit, rollback and savepoint of TCL commands

Sno.	Rollback	Commit	Savepoint
1.	Rollback means the database is restored to the last committed state	DML commands saves	Savepoint helps to save the transaction temporarily.

Sno.	Rollback	Commit	Savepoint
		modification and it permanently saves the transaction.	
2.	Syntax- ROLLBACK [To SAVEPOINT_NAME];	Syntax- COMMIT;	Syntax- SAVEPOINT [savepoint_name;]
3.	Example- ROLLBACK Update5;	Example- SQL> COMMIT;	Example- SAVEPOINT table_create;